

RAGUL NARAYANAN MAGESH

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EDUCATION

Northeastern University, Boston, MA

Sep 2024 - May 2026

Master of Science in Data Analytics Engineering, GPA - 3.8 / 4.0

Relevant Coursework: Neural Networks & Deep Learning, NLP, Machine Learning Operations, Data Mining, Generative AI

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, PySpark, TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face Transformers

Generative AI & LLMs: FLAN-T5, OpenAI API, LangChain, Prompt Engineering (zero/few-shot), RAG, Fine-tuning

MLOps: MLflow, DVC, Airflow, Docker, CI/CD, GCP (BigQuery, Vertex AI), AWS (SageMaker, S3), Experiment Tracking

NLP: BERT, RoBERTa, BM25, Vector Embeddings, Cross-Encoder Reranking, Sentiment Analysis

Data Engineering: Snowflake, PostgreSQL, MongoDB, Star Schema, ETL/ELT Pipelines | **Visualization:** Tableau, Power BI

PROJECTS

FIFA World Cup Match Predictor and Analyst Chatbot [*Langchain, OpenAI, Pinecone, Python, Streamlit, RAG*] Feb 2026

- Built a **retrieval-augmented generation (RAG) pipeline** using **LangChain**, **Pinecone vector store**, and **OpenAI embeddings** to ingest and semantically index comprehensive FIFA World Cup match data from Kaggle
- Implemented a **dual-mode chatbot** with natural language Q&A via **RAG retrieval chain** and a match prediction chain that generates sportswriter-style previews using head-to-head records and recent team form

Custom Object Detection using YOLOv8 OBB [*YOLOv8, Roboflow, PyTorch, Annotation, Data Augmentation*] Jan 2026

- Built a **YOLOv8 oriented bounding box (OBB) detection model** for 39-class object detection on 3,900+ custom-annotated images using Roboflow, applying multi-stage data augmentation and training on **NVIDIA A100 GPU**
- Validated real-world multi-object detection performance on custom collage test images, demonstrating robust generalization across complex, densely packed scenes

EchoAI - Scalable LLM-Powered Review Response System [*GCP, Airflow, BigQuery, MLflow, FLAN-T5, MLOps*] Dec 2025

- Architected and deployed an **end-to-end MLOps pipeline** processing 100K+ customer reviews using FLAN-T5 LLM for generative AI-powered automated response generation with **85%+ confidence scores**, integrated sentiment analysis, bias detection, and anomaly detection modules
- **Built production-grade ML infrastructure on GCP** with Airflow orchestration, BigQuery data warehousing, MLflow experiment tracking, DVC versioning, and CI/CD-driven deployment to a Streamlit inference application

Amazon Electronics Product Recommendation System [*PyTorch, NLP, AWS SageMaker, FastAPI, Qdrant, GCP*] Dec 2025

- Engineered a **hybrid recommendation system** processing 6.7M+ reviews across 31,100 products using BM25 sparse retrieval combined with BGE dense vector embeddings and cross-encoder reranking, achieving NDCG@10 of 0.854
- Deployed **Pegasus abstractive summarization** and **RoBERTa sentiment analysis** on AWS SageMaker; served end-to-end pipeline via FastAPI with Qdrant vector store for scalable semantic search

WORK EXPERIENCE

Data Scientist

Aug 2023 - Jul 2024

360DigiTMG

Chennai, India

- Analyzed and developed **predictive models** on large-scale IQVIA pharmaceutical datasets to forecast sales and classify high-value prescribers, performing **cohort analysis** and HCP profiling, achieving a **25% increase in targeting precision**
- Modeled and transformed large-scale datasets in Snowflake using advanced SQL, **improving reporting efficiency by 40%** and enabling seamless integration with Tableau dashboards for real-time decision-making
- Built interactive **Tableau dashboards** to visualize sales trends, prescription activity (NRx & TRx), and regional KPIs, presenting **business intelligence** insights to senior management and cross-functional leadership

Data Engineer

Apr 2022 - Jul 2023

Wipro

Bengaluru, India

- Migrated **150+ Teradata objects** containing retail, customer, and supply-chain data into **Snowflake**, performing data structuring, schema validation, and **query optimization** to ensure consistency across **disparate sources**
- Optimized ETL workflows using **PySpark** and **Apache Airflow**, cutting processing time by **30%** and automating incremental data loads across large-scale structured and unstructured datasets from multiple systems
- Built **Python-based monitoring tools** implementing **anomaly detection** and real-time alerting to enforce data quality business rules, anticipating and mitigating risks across production data infrastructure